CONTACT INFORMATION

JACKET DEMBY'S

Research Interests

- Robotics
- Machine Learning
- Deep Learning

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SUMMARY

First year Ph.D. in Electrical and Computer Engineering with growing interests in Robotics, Machine Learning and Deep Learning. Seeking research/industry experiences in Machine Learning and Deep Learning applied in Robotics applications.

EDUCATION

Master of Science Present GPA: 3.696/4	University of Missouri, Columbia, USA **Computer Engineering** Expected graduation: May 2020
Engineering Diploma Graduated July 2015 GPA: 15.06/20	Ucac-Icam Institute, Douala, Cameroon **Generalist Engineering** Top of promotion

RESEARCH EXPERIENCE

January 2020 – Present: Research Assistant (Vision Guided and Intelligent Robotics Lab – ViGIR Lab)

- **A study on the approximate solutions provided by data-driven and numerical inverse kinematics solvers for serial robotic manipulators.**
- **Choosing the correct generalized inverse for the numerical solution of the inverse kinematics of incommensurate robotic manipulators.**

July 2019 – Present: Graduate Student

- **Object detection and pose estimation using CNN in embedded hardware for assistive technology:** I am involved on a research path related to object detection and pose estimation based on stereo vision and deep learning algorithms to be used on embedded devices for affordable assistive technology.

December 2018 – Present: Graduate Student

- **Inverse kinematics of serial robotics arms using soft-computing methods:** I am involved in looking at the inverse kinematics results' accuracies provided by two soft-computing methods: artificial neural network and fuzzy neural network.

August 2018 – December 2018: Graduate Student

- **Inverse kinematics of serial robotics arms using analytical and hybrid methods:** I was involved in looking at the effects of the unit choices for robot joints in the computation of the inverse kinematics using analytical and hybrid methods.

TEACHING EXPERIENCE

Teaching Assistant, University of Missouri-Columbia, Columbia, Missouri, United-States

- **Spring 2020: Neural Networks**
- **Fall 2020: Building Intelligent Vision-Guided Robots**

PUBLICATIONS

1. **J. Demby's**, Y. Gao and G. N. DeSouza, "A Study on Solving the Inverse Kinematics of Serial Robots using Artificial Neural Network and Fuzzy Neural Network," *in 2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE),* New Orleans, LA, USA, 2019, pp. 1-6.
2. **J. Demby's**, Y. Gao, A. Shafiekhani, G.N. DeSouza, "Object Detection and Pose Estimation Using CNN in Embedded Hardware for Assistive Technology, " *in 2019 IEEE Symposium Series on Computational Intelligence (IEEE SSCI 2019)*, December 2019 (**accepted**)

RELEVANT COURSEWORK

- Introduction to Mechatronic and Robotic Vision
- Probability and Stochastic Processes for Engineers
- Introduction to Machine Learning and Pattern Recognition
- Parallel Programming for High Performance Computing
- Introduction to Computational Intelligence
- Digital Image Processing
- Computer Vision
- Supervised Learning
- Neural Networks and Deep Learning
- Real-Time Embedded Systems

SOME COURSEWORK PROJECTS INVOLVING FINAL PRESENTATIONS

Class: Neural Networks (Spring 2019)
Project: Solving the inverse kinematics of a 4 DoF SCARA robot using extreme learning machines and radial basis function networks
Main objective, results and tools: Conduct a comparative study between extreme learning machines and radial basis function networks in solving the inverse kinematics of the chosen robotic arm. Lowest MSE (Mean-Squared Error) results were obtained with extreme learning machines. Matlab and Python were used.

Class: Introduction to Computational Intelligence (Fall 2018)
Project: Solving the inverse kinematics of a 6 DoF Kinova robot using artificial and fuzzy neural networks
Main objective, results and tools: Conduct a comparative study between artificial and fuzzy neural networks in solving the inverse kinematics of the chosen robotic arm. A trade-off was observed in terms of precision of the predicted joint configuration depending on the workspace. Matlab and Python were used.

Class: Computer Vision (Spring 2018)
Project: Hardware integration of a real time object detection CNN (Yolo V2)
Main objective, results and tools: Develop an android mobile application based on YOLO V2 to recognize and pronounce objects in real-time. Android, Darknet and a pretrained Yolo V2 model were used.

Class: Introduction to Mechatronic and Robotic Vision (Fall 2017)
Project: Investigating the control of a robotic arm with a real-time tracking of human hand motions
Main objective, results and tools: Track human hand motions using a Kinect sensor and control the robotic arm with the hand motions tracking information. C/C++ and ROS were used.

Class: Digital Image Processing (Fall 2017)
Project: Super Image Restoration (SIR)
Main objective, results and tools: Develop a Matlab application with a friendly Graphical User Interface (GUI) to restore old damaged images with cracks in only three click steps. Matlab, Python and Algorithmia Colorization API were used.

Achievements and Activities

May 2019 – present	Student Member of IEEE.
December 2017	Best 3rd final project in Digital Image Processing class (Fall 2017).
August 2017 – present	Member of the University of Missouri Fulbright Organization (**MUFO**).
July 2017 – present	Recipient of the **Fulbright Scholarship** in Congo Brazzaville to pursue a M.Sc. in Computer Engineering at the University of Missouri-Columbia.
January 2017	Participant for the Business and Entrepreneurship Track of the Young African Leaders Initiative (YALI Program) in Nairobi, Kenya.
April 2016	2nd prize ($13,000) of the 2015-2016 national Startup Challenge of Total E&P in Congo Brazzaville.
January 2015 – June 2015	Participant of the internal Icam exchange program to conduct an industrial mission on Computational Fluid Dynamics (CFD) in Lille, France.
June 2012 – July 2013	Recipient of the **Sogea-Satom scholarship** to study at Ucac-Icam Cameroon.
August 2010 – June 2012	Recipient of the **Total E&P scholarship** to study at Ucac-Icam Congo.

Technologies

Operating Systems: Linux, Windows, Mac
Programming languages: C/C++, Python, MATLAB, Android, Cuda, Lyx/Latex
Software and libraries:

- **Robotics**: ROS (Robot Operating System)
- **Machine learning**: TensorFlow, Keras
- **Computer vision**: OpenCV
- **Mechanical**: Solidworks, Ansys Fluent

Work Experience

Junior Project Engineer | December 2016 – July 2017
Total E&P Congo - Pointe-Noire, Republic of Congo

- Prepared, supervised and generated daily reports for construction activities of the Floating Production Unit (FPU) Likouf alongside contracted companies.
- Worked on Pressure Safety Valves (PSV) Recalibration Campaign, managed a database of the PSV joint stocks, and ensured team members compliance with on-site safety measures.

Maintenance Engineer | September 2015 – October 2016
Sogea-Satom - Brazzaville, Republic of Congo

- Monitored the computer-assisted supply chain and maintenance management software and produced monthly analysis reports.
- Developed and maintained Excel VBA (Visual Basic for Applications) programs for monthly stocks' management analysis, optimization and periodic inventories follow up.

Intern Maintenance Engineer by Alternance | January 2013 – December 2014
Sogea-Satom - Brazzaville, Republic of Congo

- Monitored the computer-assisted supply chain and maintenance management software.
- Continually improved methods and procedures for processes, stocks' management, documenting and workflow techniques.

Intern Maintenance Technician | January 2012 – March 2012

Total E&P Congo – Pointe-Noire, Republic of Congo

- Determine the causes of cascade regulations of three-phase separators of Katina treatment unit at Djeno Oil Terminal and propose improvement solutions to ameliorate their control/regulation cycles.
- Conduct a study on the appropriate type of sensors (ultrasonic sensors to detect water, gas and oil levels) to be replaced on those separators and suggest vendors.

Intern Operator | June 2011 – July 2011

Apave Congo – Pointe-Noire, Republic of Congo

- Certify welding structures (appliances and lifting gears) using radiography, dye penetrant and magnetic particles methods.

LANGUAGES AND INTERESTS

Languages

- French: native language
- English: fluent (speaking, reading, writing)

Interests

Sport: Walking, Running, Volleyball, Swimming

Board games: Scrabble, Ludo, Chess, Checkers

VOLUNTEER WORK

Intervarsity at University of Missouri-Columbia | Volunteer photographer and events' organizer

- Organize events (games, discoveries, English clubs, shopping, etc.) to help international students feel welcome at the university.
- Take pictures during the events and share them with all the members.

University of Missouri Fulbright Organization (MUFO) | Volunteer

- Work with the International Center to provide rides to new international sponsored students.
- Lead scavenger hunt activities to help international sponsored students discover and locate key facilities and resources on campus.

OVA (Œuvre de Vie et d'Amour) Association in Cameroon | Volunteer

- Organize activities on the International Day of the African Child and Christmas for street children and children with special needs.